Newman-Penrose Formalism Calculations for

metric:

Name: Stephani et al., Equation (13.57)

Reference(s): [1]

Line element:

$$ds^2 = 2\,k^2 W\left(x_0\right)^n \cosh\left(x_0\right) \sinh\left(x_0\right) \mathrm{d}x_0 \otimes \mathrm{d}x_0 - 2\,k^2 W\left(x_0\right)^n \cosh\left(x_0\right) \sinh\left(x_0\right) \mathrm{d}x_1 \otimes \mathrm{d}x_1 + 2\,k^2 W\left(x_0\right) \cosh\left(x_0\right) e^{\left(2\,n x_1 + 2\,x_1\right)} \sinh\left(x_0\right) \mathrm{d}x_2 \otimes \mathrm{d}x_2 + \frac{2\,k^2 \cosh\left(x_0\right) e^{\left(-2\,n x_1 + 2\,x_1\right)} \sinh\left(x_0\right)}{W\left(x_0\right)} \mathrm{d}x_3 \otimes \mathrm{d}x_3 \tag{1}$$

Coordinates: (x_0, x_1, x_2, x_3)

Null tetrad components:

Covariant components:

$$l = \left(\frac{1}{2}\,\sqrt{2}\sqrt{-W\left(x_0\right)^n \sinh\left(2\,x_0\right)}k\right) dx_0 + \left(\frac{1}{2}\,\sqrt{2}\sqrt{-W\left(x_0\right)^n \sinh\left(2\,x_0\right)}k\right) dx_1 + \left(0\right) dx_2 + \left(0\right) dx_3 \tag{2}$$

$$n = \left(\frac{1}{2}\,\sqrt{2}\sqrt{-W\left(x_0\right)^n \sinh\left(2\,x_0\right)}k\right) dx_0 + \left(-\frac{1}{2}\,\sqrt{2}\sqrt{-W\left(x_0\right)^n \sinh\left(2\,x_0\right)}k\right) dx_1 + \left(0\right) dx_2 + \left(0\right) dx_3 \tag{3}$$

$$m = \left(0\right) dx_0 + \left(0\right) dx_1 + \left(\frac{1}{2}\,\sqrt{2}k\sqrt{W\left(x_0\right)}e^{\left(\left(n+1\right)x_1\right)}\sqrt{\sinh\left(2\,x_0\right)}\right) dx_2 + \left(\frac{i\,\sqrt{2}ke^{\left(-\left(n-1\right)x_1\right)}\sqrt{\sinh\left(2\,x_0\right)}}{2\,\sqrt{W\left(x_0\right)}}\right) dx_3 \tag{4}$$

$$\overline{m} = \left(0\right) dx_0 + \left(0\right) dx_1 + \left(\frac{1}{2}\,\sqrt{2}k\sqrt{W\left(x_0\right)}e^{\left(\left(n+1\right)x_1\right)}\sqrt{\sinh\left(2\,x_0\right)}\right) dx_2 + \left(-\frac{i\,\sqrt{2}ke^{\left(-\left(n-1\right)x_1\right)}\sqrt{\sinh\left(2\,x_0\right)}}{2\,\sqrt{W\left(x_0\right)}}\right) dx_3 \tag{5}$$

Contravariant components:

$$l = \left(\frac{\sqrt{-\sinh\left(x_0\right)}}{2\,k\sqrt{W\left(x_0\right)^n}\sqrt{\cosh\left(x_0\right)}\sinh\left(x_0\right)} \right) dx_0 + \left(-\frac{\sqrt{-\sinh\left(x_0\right)}}{2\,k\sqrt{W\left(x_0\right)^n}\sqrt{\cosh\left(x_0\right)}\sinh\left(x_0\right)} \right) dx_1 + (0)\,dx_2 + (0)\,dx_3 \tag{6}$$

$$n = \left(\frac{\sqrt{-\sinh\left(x_0\right)}}{2\,k\sqrt{W\left(x_0\right)^n}\sqrt{\cosh\left(x_0\right)}\sinh\left(x_0\right)} \right) dx_0 + \left(\frac{\sqrt{-\sinh\left(x_0\right)}}{2\,k\sqrt{W\left(x_0\right)^n}\sqrt{\cosh\left(x_0\right)}\sinh\left(x_0\right)} \right) dx_1 + (0)\,dx_2 + (0)\,dx_3 \tag{7}$$

$$m = (0)\,dx_0 + (0)\,dx_1 + \left(\frac{e^{(-nx_1-x_1)}}{2\,k\sqrt{W\left(x_0\right)}\sqrt{\cosh\left(x_0\right)}\sqrt{\sinh\left(x_0\right)}} \right) dx_2 + \left(\frac{i\,\sqrt{W\left(x_0\right)}e^{(nx_1-x_1)}}{2\,k\sqrt{\cosh\left(x_0\right)}\sqrt{\sinh\left(x_0\right)}} \right) dx_3 \tag{8}$$

$$\overline{m} = (0)\,dx_0 + (0)\,dx_1 + \left(\frac{e^{(-nx_1-x_1)}}{2\,k\sqrt{W\left(x_0\right)}\sqrt{\cosh\left(x_0\right)}\sqrt{\sinh\left(x_0\right)}} \right) dx_2 + \left(-\frac{i\,\sqrt{W\left(x_0\right)}e^{(nx_1-x_1)}}{2\,k\sqrt{\cosh\left(x_0\right)}\sqrt{\sinh\left(x_0\right)}} \right) dx_3 \tag{9}$$

Directional derivatives:

$$D = \frac{\sqrt{-\sinh(x_0)}\,\frac{\partial}{\partial x_0} f(x_0, x_1, x_2, x_3)}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{\sqrt{-\sinh(x_0)}\,\frac{\partial}{\partial x_1} f(x_0, x_1, x_2, x_3)}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} \tag{10}$$

$$\Delta = \frac{\sqrt{-\sinh(x_0)}\,\frac{\partial}{\partial x_0} f(x_0, x_1, x_2, x_3)}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{\sqrt{-\sinh(x_0)}\,\frac{\partial}{\partial x_1} f(x_0, x_1, x_2, x_3)}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} \tag{11}$$

$$\delta = \frac{i\sqrt{W(x_0)}e^{(nx_1 - x_1)}\frac{\partial}{\partial x_3} f(x_0, x_1, x_2, x_3)}{2\,k\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{e^{(-nx_1 - x_1)}\frac{\partial}{\partial x_2} f(x_0, x_1, x_2, x_3)}{2\,k\sqrt{W(x_0)}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} \tag{12}$$

$$\overline{\delta} = -\frac{i\sqrt{W(x_0)}e^{(nx_1 - x_1)}\frac{\partial}{\partial x_3} f(x_0, x_1, x_2, x_3)}{2\,k\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} + \frac{e^{(-nx_1 - x_1)}\frac{\partial}{\partial x_2} f(x_0, x_1, x_2, x_3)}{2\,k\sqrt{W(x_0)}\sqrt{\cosh(x_0)}\sqrt{\sinh(x_0)}} \tag{13}$$

Spin coefficients:

$$\rho = -\frac{\sqrt{-\sinh(x_0)}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2} + \frac{\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{14}$$

$$\sigma = \frac{n\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} \tag{15}$$

$$\kappa = 0 \tag{16}$$

$$\mu = \frac{\sqrt{-\sinh(x_0)}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2} + \frac{\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{17}$$

$$\lambda = \frac{\left(2\,nW(x_0) + \frac{\partial}{\partial x_0}W(x_0)\right)\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} \tag{18}$$

$$\nu = 0 \tag{19}$$

$$\alpha = 0 \tag{20}$$

$$\beta = 0 \tag{21}$$

$$\tau = 0 \tag{22}$$

$$\pi = 0 \tag{23}$$

$$\epsilon = \frac{n\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{8\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} + \frac{\sqrt{-\sinh(x_0)}}{8\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{24}$$

$$\gamma = -\frac{n\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{8\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{\sqrt{-\sinh(x_0)}}{8\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{25}$$

Relations between spin coefficients:

Ricci tensor components:

$$\Phi_{00} = \frac{n^2 W(x_0)^{-n} \cosh(x_0)}{4\,k^2 \sinh(x_0)^3} - \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \sinh(x_0)^2} - \frac{n^2 W(x_0)^{-n}}{4\,k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3} + \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{3\,W(x_0)^{-n}}{16\,k^2 \cosh(x_0)^3 \sinh(x_0)^3} \tag{26}$$

$$\Phi_{01} = 0 \tag{27}$$

$$\Phi_{02} = \frac{n W(x_0)^{-n}}{2\,k^2 \cosh(x_0) \sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \cosh(x_0)^2} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0) \sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} \tag{28}$$

$$\Phi_{10} = 0 \tag{29}$$

$$\Phi_{11} = -\frac{n^2 W(x_0)^{-n} \cosh(x_0)}{8\,k^2 \sinh(x_0)^3} - \frac{n W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3} + \frac{n W(x_0)^{-n} \cosh(x_0) \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{16\,k^2 W(x_0) \sinh(x_0)^3} + \frac{n^2 W(x_0)^{-n}}{8\,k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{32\,k^2 W(x_0)^2 \sinh(x_0)^3} + \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{n W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{16\,k^2 W(x_0) \cosh(x_0) \sinh(x_0)^3} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{32\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{3\,W(x_0)^{-n}}{32\,k^2 \cosh(x_0)^3 \sinh(x_0)^3} \tag{30}$$

$$\Phi_{12} = 0 \tag{31}$$

$$\Phi_{20} = \frac{nW\left(x_0\right)^{-n}}{2\,k^2\cosh\left(x_0\right)\sinh\left(x_0\right)} - \frac{W\left(x_0\right)^{-n}\frac{\partial}{\partial x_0}W\left(x_0\right)}{4\,k^2W\left(x_0\right)\cosh\left(x_0\right)^2} + \frac{W\left(x_0\right)^{-n}\frac{\partial}{\partial x_0}W\left(x_0\right)^2}{8\,k^2W\left(x_0\right)^2\cosh\left(x_0\right)\sinh\left(x_0\right)} - \frac{W\left(x_0\right)^{-n}\frac{\partial^2}{(\partial x_0)^2}W\left(x_0\right)}{8\,k^2W\left(x_0\right)\cosh\left(x_0\right)\sinh\left(x_0\right)} - \frac{W\left(x_0\right)^{-n}\frac{\partial}{\partial x_0}W\left(x_0\right)}{8\,k^2W\left(x_0\right)\cosh\left(x_0\right)^2\sinh\left(x_0\right)^2} \tag{32}$$

$$\Phi_{21} = 0 \tag{33}$$

$$\begin{aligned}\Phi_{22} = {} & \frac{n^2W\left(x_0\right)^{-n}\cosh\left(x_0\right)}{4\,k^2\sinh\left(x_0\right)^3} - \frac{nW\left(x_0\right)^{-n}\frac{\partial}{\partial x_0}W\left(x_0\right)}{4\,k^2W\left(x_0\right)\sinh\left(x_0\right)^2} - \frac{n^2W\left(x_0\right)^{-n}}{4\,k^2\cosh\left(x_0\right)\sinh\left(x_0\right)^3} + \frac{W\left(x_0\right)^{-n}\cosh\left(x_0\right)\frac{\partial}{\partial x_0}W\left(x_0\right)^2}{16\,k^2W\left(x_0\right)^2\sinh\left(x_0\right)^3} \\ & + \frac{nW\left(x_0\right)^{-n}\frac{\partial}{\partial x_0}W\left(x_0\right)}{8\,k^2W\left(x_0\right)\cosh\left(x_0\right)^2\sinh\left(x_0\right)^2} - \frac{W\left(x_0\right)^{-n}\frac{\partial}{\partial x_0}W\left(x_0\right)^2}{16\,k^2W\left(x_0\right)^2\cosh\left(x_0\right)\sinh\left(x_0\right)^3} - \frac{3\,W\left(x_0\right)^{-n}}{16\,k^2\cosh\left(x_0\right)^3\sinh\left(x_0\right)^3}\end{aligned} \tag{34}$$

$$\begin{aligned}\Lambda = {} & \frac{n^2W\left(x_0\right)^{-n}\cosh\left(x_0\right)}{24\,k^2\sinh\left(x_0\right)^3} + \frac{nW\left(x_0\right)^{-n}\cosh\left(x_0\right)\frac{\partial}{\partial x_0}W\left(x_0\right)^2}{48\,k^2W\left(x_0\right)^2\sinh\left(x_0\right)^3} - \frac{nW\left(x_0\right)^{-n}\cosh\left(x_0\right)\frac{\partial^2}{(\partial x_0)^2}W\left(x_0\right)}{48\,k^2W\left(x_0\right)\sinh\left(x_0\right)^3} \\ & - \frac{n^2W\left(x_0\right)^{-n}}{24\,k^2\cosh\left(x_0\right)\sinh\left(x_0\right)^3} - \frac{W\left(x_0\right)^{-n}\cosh\left(x_0\right)\frac{\partial}{\partial x_0}W\left(x_0\right)^2}{96\,k^2W\left(x_0\right)^2\sinh\left(x_0\right)^3} - \frac{nW\left(x_0\right)^{-n}\frac{\partial}{\partial x_0}W\left(x_0\right)^2}{48\,k^2W\left(x_0\right)^2\cosh\left(x_0\right)\sinh\left(x_0\right)^3} \\ & + \frac{nW\left(x_0\right)^{-n}\frac{\partial^2}{(\partial x_0)^2}W\left(x_0\right)}{48\,k^2W\left(x_0\right)\cosh\left(x_0\right)\sinh\left(x_0\right)^3} + \frac{W\left(x_0\right)^{-n}\frac{\partial}{\partial x_0}W\left(x_0\right)^2}{96\,k^2W\left(x_0\right)^2\cosh\left(x_0\right)\sinh\left(x_0\right)^3} + \frac{W\left(x_0\right)^{-n}}{32\,k^2\cosh\left(x_0\right)^3\sinh\left(x_0\right)^3}\end{aligned} \tag{35}$$

Weyl tensor components:

$$\Psi_0 = \frac{n^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} - \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \tag{36}$$

$$\Psi_1 = 0 \tag{37}$$

$$\Psi_2 = \frac{n^2 W(x_0)^{-n}}{6\,k^2\cosh(x_0)\sinh(x_0)} - \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{24\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{n W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{24\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{24\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} \tag{38}$$

$$\Psi_3 = 0 \tag{39}$$

$$\Psi_4 = -\frac{n^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} - \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \tag{40}$$

Newman-Penrose equations:

$$NP1 = 0 \tag{41}$$

$$NP2 = 0 \tag{42}$$

$$NP3 = 0 \tag{43}$$

$$NP4 = 0 \tag{44}$$

$$NP5 = 0 \tag{45}$$

$$NP6 = 0 \tag{46}$$

$$NP7 = 0 \tag{47}$$

$$NP8 = 0 \tag{48}$$

$$NP9 = 0 \tag{49}$$

$$NP10 = 0 \tag{50}$$

$$NP11 = 0 \tag{51}$$

$$NP12 = 0 \tag{52}$$

$$NP13 = 0 \tag{53}$$

$$NP14 = 0 \tag{54}$$

$$NP15 = 0 \tag{55}$$

$$NP16 = 0 \tag{56}$$

$$NP17 = 0 \tag{57}$$

$$NP18 = 0 \tag{58}$$

Bianchi identities:

$$BI1 = 0 \tag{59}$$

$$BI2 = 0 \tag{60}$$

$$BI3 = 0 \tag{61}$$

$$BI4 = 0 \tag{62}$$

$$BI5 = 0 \tag{63}$$

$$BI6 = 0 \tag{64}$$

$$BI7 = 0 \tag{65}$$

$$BI8 = 0 \tag{66}$$

$$BI9 = 0 \tag{67}$$

$$BI10 = 0 \tag{68}$$

$$BI11 = 0 \tag{69}$$

Petrov classification:

Petrov class:

A Type transformations:

Spin coefficients:

$$\rho' = \rho \tag{70}$$

$$\sigma' = \sigma \tag{71}$$

$$\kappa' = 0 \tag{72}$$

$$\begin{aligned}\mu' = {} & \frac{n z^2 \sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{n z^2 \sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{z^2\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{z^2\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} \\ & - \frac{z\sqrt{-\sinh(x_0)}\sqrt{\cosh(x_0)}}{k\sqrt{W(x_0)^n}\sinh(x_0)^2} + \frac{z\sqrt{-\sinh(x_0)}}{k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{z^2\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} + \frac{\sqrt{-\sinh(x_0)}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2} \\ & + \frac{\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{z\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} - \frac{\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2}\end{aligned} \tag{73}$$

$$\lambda' = \lambda \tag{74}$$

$$\begin{aligned}\nu' = {} & \frac{n z^2 \sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{z^2\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{z^2\sqrt{-\sinh(x_0)}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2} \\ & + \frac{z^2\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{z^2\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2}\end{aligned} \tag{75}$$

$$\begin{aligned}\alpha' = {} & \frac{n z\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{8\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{z\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{z\sqrt{-\sinh(x_0)}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2} \\ & + \frac{z\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{3\,z\sqrt{-\sinh(x_0)}}{8\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2}\end{aligned} \tag{76}$$

$$\beta' = \frac{nz\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{n\sqrt{-\sinh(x_0)}\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{8\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{\sqrt{-\sinh(x_0)}\overline{z}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{z\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{\sqrt{-\sinh(x_0)}\overline{z}}{8\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{77}$$

$$\tau' = \frac{nz\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{8\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{z\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} + \frac{z\sqrt{-\sinh(x_0)}}{8\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{78}$$

$$\pi' = \frac{nz\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{z\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} \tag{79}$$

$$\epsilon' = \epsilon \tag{80}$$

$$\begin{aligned}\gamma' = {} & \frac{nz^2\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{nz\sqrt{-\sinh(x_0)}\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{8\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{z\sqrt{-\sinh(x_0)}\overline{z}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} \\ & - \frac{z^2\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{n\sqrt{-\sinh(x_0)}\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{8\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{\sqrt{-\sinh(x_0)}\overline{z}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} \\ & - \frac{\sqrt{-\sinh(x_0)}\overline{z}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2} + \frac{\sqrt{-\sinh(x_0)}\overline{z}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{n\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{8\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} \\ & + \frac{z\sqrt{-\sinh(x_0)}\overline{z}}{8\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} + \frac{3\sqrt{-\sinh(x_0)}\overline{z}}{8\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} - \frac{\sqrt{-\sinh(x_0)}}{8\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2}\end{aligned} \tag{81}$$

Weyl components:

$$\Psi_0' = \Psi_0 \tag{82}$$

$$\Psi_1' = \frac{n^2 z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n z W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} - \frac{n z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)}$$
$$- \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{z W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \tag{83}$$

$$\Psi_2' = \frac{n^2 z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n z^2 W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} - \frac{n z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{n^2 W(x_0)^{-n}}{6\,k^2\cosh(x_0)\sinh(x_0)}$$
$$- \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{z^2 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \tag{84}$$
$$- \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{24\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{n W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{24\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{24\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)}$$

$$\Psi_3' = \frac{n^2 z^3 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n z^3 W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} - \frac{n z^3 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{n^2 z W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)}$$
$$- \frac{z^3 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z^3 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{z^3 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \tag{85}$$
$$- \frac{n z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{n z W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)}$$

$$\Psi_4' = \frac{n^2 z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n z^4 W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} - \frac{n z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{n^2 z^2 W(x_0)^{-n}}{k^2\cosh(x_0)\sinh(x_0)}$$
$$- \frac{z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{z^4 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{n z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{4\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} \tag{86}$$
$$+ \frac{n z^2 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{n^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{4\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{n W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)}$$
$$- \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)}$$

Ricci components:

$$\Phi'_{00} = \Phi_{00} \tag{87}$$

$$\begin{aligned}\Phi'_{01} = {} & \frac{n^2 z W(x_0)^{-n} \cosh(x_0)}{4\,k^2 \sinh(x_0)^3} - \frac{n z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \sinh(x_0)^2} - \frac{n^2 z W(x_0)^{-n}}{4\,k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{z W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3} \\ & + \frac{n z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{3\,z W(x_0)^{-n}}{16\,k^2 \cosh(x_0)^3 \sinh(x_0)^3}\end{aligned} \tag{88}$$

$$\begin{aligned}\Phi'_{02} = {} & \frac{n^2 z^2 W(x_0)^{-n} \cosh(x_0)}{4\,k^2 \sinh(x_0)^3} - \frac{n z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \sinh(x_0)^2} - \frac{n^2 z^2 W(x_0)^{-n}}{4\,k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{z^2 W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3} \\ & + \frac{n W(x_0)^{-n}}{2\,k^2 \cosh(x_0) \sinh(x_0)} + \frac{n z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \cosh(x_0)^2} - \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} \\ & + \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0) \sinh(x_0)} - \frac{3\,z^2 W(x_0)^{-n}}{16\,k^2 \cosh(x_0)^3 \sinh(x_0)^3} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2}\end{aligned} \tag{89}$$

$$\begin{aligned}\Phi'_{10} = {} & \frac{n^2 W(x_0)^{-n} \overline{z} \cosh(x_0)}{4\,k^2 \sinh(x_0)^3} - \frac{n W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \sinh(x_0)^2} - \frac{n^2 W(x_0)^{-n} \overline{z}}{4\,k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3} \\ & + \frac{n W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{3\,W(x_0)^{-n} \overline{z}}{16\,k^2 \cosh(x_0)^3 \sinh(x_0)^3}\end{aligned} \tag{90}$$

$$\begin{aligned}\Phi'_{11} = {} & \frac{n^2 z W(x_0)^{-n} \overline{z} \cosh(x_0)}{4\,k^2 \sinh(x_0)^3} - \frac{n z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \sinh(x_0)^2} - \frac{n^2 z W(x_0)^{-n} \overline{z}}{4\,k^2 \cosh(x_0) \sinh(x_0)^3} - \frac{n^2 W(x_0)^{-n} \cosh(x_0)}{8\,k^2 \sinh(x_0)^3} + \frac{z W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3} \\ & + \frac{n z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{n W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3} + \frac{n W(x_0)^{-n} \cosh(x_0) \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{16\,k^2 W(x_0) \sinh(x_0)^3} \\ & + \frac{n^2 W(x_0)^{-n}}{8\,k^2 \cosh(x_0) \sinh(x_0)^3} - \frac{z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{32\,k^2 W(x_0)^2 \sinh(x_0)^3} + \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} \\ & - \frac{n W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{16\,k^2 W(x_0) \cosh(x_0) \sinh(x_0)^3} - \frac{3\,z W(x_0)^{-n} \overline{z}}{16\,k^2 \cosh(x_0)^3 \sinh(x_0)^3} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{32\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{3\,W(x_0)^{-n}}{32\,k^2 \cosh(x_0)^3 \sinh(x_0)^3}\end{aligned} \tag{91}$$

$$
\begin{aligned}
\Phi'_{12} = {} & \frac{n^2 z^2 W(x_0)^{-n}\,\overline{z}\cosh(x_0)}{4\,k^2\sinh(x_0)^3} - \frac{n z^2 W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0}W(x_0)}{4\,k^2 W(x_0)\sinh(x_0)^2} - \frac{n^2 z^2 W(x_0)^{-n}\,\overline{z}}{4\,k^2\cosh(x_0)\sinh(x_0)^3} - \frac{n^2 z W(x_0)^{-n}\cosh(x_0)}{4\,k^2\sinh(x_0)^3} + \frac{z^2 W(x_0)^{-n}\,\overline{z}\cosh(x_0)\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\sinh(x_0)^3} \\
& + \frac{n W(x_0)^{-n}\,\overline{z}}{2\,k^2\cosh(x_0)\sinh(x_0)} + \frac{n z^2 W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} - \frac{n z W(x_0)^{-n}\cosh(x_0)\,\frac{\partial}{\partial x_0}W(x_0)^2}{8\,k^2 W(x_0)^2\sinh(x_0)^3} + \frac{n z W(x_0)^{-n}\cosh(x_0)\,\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,k^2 W(x_0)\sinh(x_0)^3} \\
& + \frac{n^2 z W(x_0)^{-n}}{4\,k^2\cosh(x_0)\sinh(x_0)^3} - \frac{W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0}W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)^2} - \frac{z^2 W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} + \frac{z W(x_0)^{-n}\cosh(x_0)\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\sinh(x_0)^3} \\
& + \frac{W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0}W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n}\,\overline{z}\,\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n z W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} - \frac{n z W(x_0)^{-n}\,\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)^3} \\
& - \frac{3\,z^2 W(x_0)^{-n}\,\overline{z}}{16\,k^2\cosh(x_0)^3\sinh(x_0)^3} - \frac{W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} - \frac{z W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} - \frac{3\,z W(x_0)^{-n}}{16\,k^2\cosh(x_0)^3\sinh(x_0)^3}
\end{aligned}
\tag{92}
$$

$$
\begin{aligned}
\Phi'_{20} = {} & \frac{n^2 W(x_0)^{-n}\,\overline{z}^2\cosh(x_0)}{4\,k^2\sinh(x_0)^3} - \frac{n W(x_0)^{-n}\,\overline{z}^2\,\frac{\partial}{\partial x_0}W(x_0)}{4\,k^2 W(x_0)\sinh(x_0)^2} - \frac{n^2 W(x_0)^{-n}\,\overline{z}^2}{4\,k^2\cosh(x_0)\sinh(x_0)^3} + \frac{W(x_0)^{-n}\,\overline{z}^2\cosh(x_0)\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\sinh(x_0)^3} \\
& + \frac{n W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} + \frac{n W(x_0)^{-n}\,\overline{z}^2\,\frac{\partial}{\partial x_0}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} - \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)^2} - \frac{W(x_0)^{-n}\,\overline{z}^2\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} \\
& + \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n}\,\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{3\,W(x_0)^{-n}\,\overline{z}^2}{16\,k^2\cosh(x_0)^3\sinh(x_0)^3} - \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2}
\end{aligned}
\tag{93}
$$

$$
\begin{aligned}
\Phi'_{21} = {} & \frac{n^2 z W(x_0)^{-n} \overline{z}^2 \cosh(x_0)}{4\,k^2 \sinh(x_0)^3} - \frac{n z W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \sinh(x_0)^2} - \frac{n^2 z W(x_0)^{-n} \overline{z}^2}{4\,k^2 \cosh(x_0) \sinh(x_0)^3} - \frac{n^2 W(x_0)^{-n} \overline{z} \cosh(x_0)}{4\,k^2 \sinh(x_0)^3} + \frac{z W(x_0)^{-n} \overline{z}^2 \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3} \\
& + \frac{n z W(x_0)^{-n}}{2\,k^2 \cosh(x_0) \sinh(x_0)} + \frac{n z W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{n W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2 \sinh(x_0)^3} + \frac{n W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0) \sinh(x_0)^3} \\
& + \frac{n^2 W(x_0)^{-n} \overline{z}}{4\,k^2 \cosh(x_0) \sinh(x_0)^3} - \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \cosh(x_0)^2} - \frac{z W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3} \\
& + \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} - \frac{z W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0) \sinh(x_0)} + \frac{n W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{n W(x_0)^{-n} \overline{z} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0) \sinh(x_0)^3} \\
& - \frac{3\,z W(x_0)^{-n} \overline{z}^2}{16\,k^2 \cosh(x_0)^3 \sinh(x_0)^3} - \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{3\,W(x_0)^{-n} \overline{z}}{16\,k^2 \cosh(x_0)^3 \sinh(x_0)^3}
\end{aligned}
\tag{94}
$$

$$
\begin{aligned}
\Phi'_{22} = {} & \frac{n^2 z^2 W(x_0)^{-n} \overline{z}^2 \cosh(x_0)}{4\,k^2 \sinh(x_0)^3} - \frac{n z^2 W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \sinh(x_0)^2} - \frac{n^2 z^2 W(x_0)^{-n} \overline{z}^2}{4\,k^2 \cosh(x_0) \sinh(x_0)^3} - \frac{n^2 z W(x_0)^{-n} \overline{z} \cosh(x_0)}{2\,k^2 \sinh(x_0)^3} + \frac{z^2 W(x_0)^{-n} \overline{z}^2 \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3} \\
& + \frac{n z^2 W(x_0)^{-n}}{2\,k^2 \cosh(x_0) \sinh(x_0)} + \frac{n W(x_0)^{-n} \overline{z}^2}{2\,k^2 \cosh(x_0) \sinh(x_0)} + \frac{n z^2 W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{n z W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{4\,k^2 W(x_0)^2 \sinh(x_0)^3} \\
& + \frac{n z W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{4\,k^2 W(x_0) \sinh(x_0)^3} + \frac{n^2 z W(x_0)^{-n} \overline{z}}{2\,k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{n^2 W(x_0)^{-n} \cosh(x_0)}{4\,k^2 \sinh(x_0)^3} - \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \cosh(x_0)^2} \\
& - \frac{W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \cosh(x_0)^2} - \frac{z^2 W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{z W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2 \sinh(x_0)^3} + \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} \\
& + \frac{W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} - \frac{z^2 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0) \sinh(x_0)} - \frac{W(x_0)^{-n} \overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0) \sinh(x_0)} - \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \sinh(x_0)^2} \\
& + \frac{n z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{4\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{n z W(x_0)^{-n} \overline{z} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{4\,k^2 W(x_0) \cosh(x_0) \sinh(x_0)^3} - \frac{3\,z^2 W(x_0)^{-n} \overline{z}^2}{16\,k^2 \cosh(x_0)^3 \sinh(x_0)^3} - \frac{n^2 W(x_0)^{-n}}{4\,k^2 \cosh(x_0) \sinh(x_0)^3} \\
& - \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3} \\
& + \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{3\,z W(x_0)^{-n} \overline{z}}{8\,k^2 \cosh(x_0)^3 \sinh(x_0)^3} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{3\,W(x_0)^{-n}}{16\,k^2 \cosh(x_0)^3 \sinh(x_0)^3}
\end{aligned}
\tag{95}
$$

$$\Lambda' = \Lambda \tag{96}$$

B Type transformations:

Spin coefficients:

$$\rho' = -\frac{z\sqrt{-\sinh(x_0)}\overline{z}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2} + \frac{z\sqrt{-\sinh(x_0)}\overline{z}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{z\sqrt{-\sinh(x_0)}\overline{z}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{97}$$

$$\sigma' = \frac{nz^2\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{z^2\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} \tag{98}$$

$$\kappa' = 0 \tag{99}$$

$$\mu' = \frac{\sqrt{-\sinh(x_0)}\sqrt{\cosh(x_0)}}{2\,kz\sqrt{W(x_0)^n}\overline{z}\sinh(x_0)^2} + \frac{\sqrt{-\sinh(x_0)}}{2\,kz\sqrt{W(x_0)^n}\overline{z}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{\sqrt{-\sinh(x_0)}}{4\,kz\sqrt{W(x_0)^n}\overline{z}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{100}$$

$$\lambda' = \frac{n\sqrt{-\sinh(x_0)}}{2\,kz^2\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,kz^2\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} \tag{101}$$

$$\nu' = 0 \tag{102}$$

$$\alpha' = 0 \tag{103}$$

$$\beta' = 0 \tag{104}$$

$$\tau' = 0 \tag{105}$$

$$\pi' = 0 \tag{106}$$

$$\epsilon' = \frac{nz\sqrt{-\sinh(x_0)}\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{8\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{z\sqrt{-\sinh(x_0)}\overline{z}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} + \frac{z\sqrt{-\sinh(x_0)}\overline{z}}{8\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{107}$$

$$\gamma' = -\frac{n\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{8\,kz\sqrt{W(x_0)^n}W(x_0)\,\overline{z}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{\sqrt{-\sinh(x_0)}}{4\,kz\sqrt{W(x_0)^n}\overline{z}\cosh(x_0)^{\frac{3}{2}}} - \frac{\sqrt{-\sinh(x_0)}}{8\,kz\sqrt{W(x_0)^n}\overline{z}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{108}$$

Weyl components:

$$\begin{aligned}\Psi_0' = {} & \frac{n^2z^4W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,k^2W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{nz^4W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} - \frac{nz^4W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{8\,k^2W(x_0)^2\cosh(x_0)\sinh(x_0)} \\ & - \frac{z^4W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)}{4\,k^2W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z^4W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{8\,k^2W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{z^4W(x_0)^{-n}\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,k^2W(x_0)\cosh(x_0)\sinh(x_0)}\end{aligned} \tag{109}$$

$$\Psi_1' = 0 \tag{110}$$

$$\Psi_2' = \frac{n^2z^2W(x_0)^{-n}}{6\,k^2\cosh(x_0)\sinh(x_0)} - \frac{nz^2W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{24\,k^2W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{nz^2W(x_0)^{-n}\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{24\,k^2W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z^2W(x_0)^{-n}\frac{\partial}{\partial x_0}W(x_0)^2}{24\,k^2W(x_0)^2\cosh(x_0)\sinh(x_0)} \tag{111}$$

$$\Psi_3' = 0 \tag{112}$$

$$\Psi_4' = \Psi_4 \tag{113}$$

Ricci components:

$$\begin{aligned}\Phi_{00}' = {} & \frac{n^2z^2W(x_0)^{-n}\overline{z}^2\cosh(x_0)}{4\,k^2\sinh(x_0)^3} - \frac{nz^2W(x_0)^{-n}\overline{z}^2\frac{\partial}{\partial x_0}W(x_0)}{4\,k^2W(x_0)\sinh(x_0)^2} - \frac{n^2z^2W(x_0)^{-n}\overline{z}^2}{4\,k^2\cosh(x_0)\sinh(x_0)^3} + \frac{z^2W(x_0)^{-n}\overline{z}^2\cosh(x_0)\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2W(x_0)^2\sinh(x_0)^3} \\ & + \frac{nz^2W(x_0)^{-n}\overline{z}^2\frac{\partial}{\partial x_0}W(x_0)}{8\,k^2W(x_0)\cosh(x_0)^2\sinh(x_0)^2} - \frac{z^2W(x_0)^{-n}\overline{z}^2\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2W(x_0)^2\cosh(x_0)\sinh(x_0)^3} - \frac{3\,z^2W(x_0)^{-n}\overline{z}^2}{16\,k^2\cosh(x_0)^3\sinh(x_0)^3}\end{aligned} \tag{114}$$

$$\Phi'_{01} = 0 \tag{115}$$

$$\Phi'_{02} = \frac{nz^2 W(x_0)^{-n}}{2\,k^2 \cosh(x_0)\sinh(x_0)} - \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)^2} + \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{z^2 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2 \sinh(x_0)^2} \tag{116}$$

$$\Phi'_{10} = 0 \tag{117}$$

$$\begin{aligned}\Phi'_{11} = &-\frac{n^2 z W(x_0)^{-n}\,\overline{z}\cosh(x_0)}{8\,k^2 \sinh(x_0)^3} - \frac{nzW(x_0)^{-n}\,\overline{z}\cosh(x_0)\frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3} + \frac{nzW(x_0)^{-n}\,\overline{z}\cosh(x_0)\frac{\partial^2}{(\partial x_0)^2} W(x_0)}{16\,k^2 W(x_0)\sinh(x_0)^3} \\ &+ \frac{n^2 z W(x_0)^{-n}\,\overline{z}}{8\,k^2 \cosh(x_0)\sinh(x_0)^3} + \frac{zW(x_0)^{-n}\,\overline{z}\cosh(x_0)\frac{\partial}{\partial x_0} W(x_0)^2}{32\,k^2 W(x_0)^2 \sinh(x_0)^3} + \frac{nzW(x_0)^{-n}\,\overline{z}\frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)^3} \\ &- \frac{nzW(x_0)^{-n}\,\overline{z}\frac{\partial^2}{(\partial x_0)^2} W(x_0)}{16\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)^3} - \frac{zW(x_0)^{-n}\,\overline{z}\frac{\partial}{\partial x_0} W(x_0)^2}{32\,k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)^3} - \frac{3\,zW(x_0)^{-n}\,\overline{z}}{32\,k^2 \cosh(x_0)^3 \sinh(x_0)^3}\end{aligned} \tag{118}$$

$$\Phi'_{12} = 0 \tag{119}$$

$$\Phi'_{20} = \frac{nW(x_0)^{-n}\,\overline{z}^2}{2\,k^2 \cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n}\,\overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)^2} + \frac{W(x_0)^{-n}\,\overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2 \cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n}\,\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n}\,\overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2 \sinh(x_0)^2} \tag{120}$$

$$\Phi'_{21} = 0 \tag{121}$$

$$\Phi'_{22} = \Phi_{22} \tag{122}$$

$$\Lambda' = \Lambda \tag{123}$$

C Type transformations:

Spin coefficients:

$$\rho' = \frac{z\sqrt{-\sinh(x_0)}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2} + \frac{\sqrt{-\sinh(x_0)}\overline{z}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2} + \frac{z\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)}$$
$$+ \frac{\sqrt{-\sinh(x_0)}\overline{z}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{\sqrt{-\sinh(x_0)}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2} + \frac{\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} \tag{124}$$
$$- \frac{z\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} - \frac{\sqrt{-\sinh(x_0)}\overline{z}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} + \frac{\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2}$$

$$\sigma' = -\frac{nz^2\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{z^2\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} + \frac{z^2\sqrt{-\sinh(x_0)}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2} + \frac{z^2\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} \tag{125}$$
$$+ \frac{n\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{z^2\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} - \frac{\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)}$$

$$\kappa' = \frac{nz^3\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{nz^2\sqrt{-\sinh(x_0)}\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{z^2\sqrt{-\sinh(x_0)}\overline{z}}{2\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}}$$
$$+ \frac{z^2\sqrt{-\sinh(x_0)}\overline{z}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2} + \frac{z^2\sqrt{-\sinh(x_0)}\overline{z}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{z^3\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} \tag{126}$$
$$+ \frac{nz\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{nz\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{z\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}}$$
$$- \frac{z^2\sqrt{-\sinh(x_0)}\overline{z}}{2\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} - \frac{z\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{z\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2}$$

$$\mu' = \frac{nz^2\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{nz\sqrt{-\sinh(x_0)}\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{z\sqrt{-\sinh(x_0)}\overline{z}}{2\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} + \frac{z^2\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)}$$
$$- \frac{z\sqrt{-\sinh(x_0)}\overline{z}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} + \frac{\sqrt{-\sinh(x_0)}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2} + \frac{\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{127}$$

$$\lambda' = \lambda \tag{128}$$

$$\nu' = 0 \tag{129}$$

$$\alpha' = -\frac{nz\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{8\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{z\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{z\sqrt{-\sinh(x_0)}}{8\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{130}$$

$$\beta' = -\frac{nz\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{8\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{z\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} + \frac{z\sqrt{-\sinh(x_0)}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2}$$
$$+ \frac{z\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{3\,z\sqrt{-\sinh(x_0)}}{8\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{131}$$

$$\tau' = -\frac{nz\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{z\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{z\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{132}$$

$$\pi' = 0 \tag{133}$$

$$\epsilon' = \frac{nz^2\sqrt{-\sinh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{nz\sqrt{-\sinh(x_0)}\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{8\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} - \frac{z\sqrt{-\sinh(x_0)}\overline{z}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}}$$
$$+ \frac{z\sqrt{-\sinh(x_0)}\overline{z}\sqrt{\cosh(x_0)}}{2\,k\sqrt{W(x_0)^n}\sinh(x_0)^2} + \frac{z\sqrt{-\sinh(x_0)}\overline{z}}{2\,k\sqrt{W(x_0)^n}\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{z^2\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{4\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)}$$
$$+ \frac{n\sqrt{-\sinh(x_0)}\frac{\partial}{\partial x_0}W(x_0)}{8\,k\sqrt{W(x_0)^n}W(x_0)\sqrt{\cosh(x_0)}\sinh(x_0)} + \frac{\sqrt{-\sinh(x_0)}}{4\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}} - \frac{3\,z\sqrt{-\sinh(x_0)}\overline{z}}{8\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} + \frac{\sqrt{-\sinh(x_0)}}{8\,k\sqrt{W(x_0)^n}\cosh(x_0)^{\frac{3}{2}}\sinh(x_0)^2} \tag{134}$$

$$\gamma' = \gamma \tag{135}$$

Weyl components:

$$\begin{aligned}\Psi_0' = &-\frac{n^2 z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n z^4 W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} - \frac{n z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{n^2 z^2 W(x_0)^{-n}}{k^2\cosh(x_0)\sinh(x_0)} \\ &+ \frac{z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z^4 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{z^4 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{n z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{4\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} \\ &+ \frac{n z^2 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{4\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{n W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} \\ &- \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)}\end{aligned} \tag{136}$$

$$\begin{aligned}\Psi_1' = &-\frac{n^2 z^3 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n z^3 W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} - \frac{n z^3 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{n^2 z W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} \\ &+ \frac{z^3 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z^3 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{z^3 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \\ &- \frac{n z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{n z W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)}\end{aligned} \tag{137}$$

$$\begin{aligned}\Psi_2' = &-\frac{n^2 z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n z^2 W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} - \frac{n z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{n^2 W(x_0)^{-n}}{6\,k^2\cosh(x_0)\sinh(x_0)} \\ &+ \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z^2 W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{z^2 W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \\ &- \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{24\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{n W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{24\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{24\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)}\end{aligned} \tag{138}$$

$$\Psi_3' = -\frac{n^2 z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n z W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} - \frac{n z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)}$$
$$+ \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} + \frac{z W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} \tag{139}$$

$$\Psi_4' = \Psi_4 \tag{140}$$

Ricci components:

$$\Phi_{00}' = \frac{n^2 z^2 W(x_0)^{-n}\,\overline{z}^2\cosh(x_0)}{4\,k^2\sinh(x_0)^3} - \frac{n z^2 W(x_0)^{-n}\,\overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\sinh(x_0)^2} - \frac{n^2 z^2 W(x_0)^{-n}\,\overline{z}^2}{4\,k^2\cosh(x_0)\sinh(x_0)^3} - \frac{n^2 z W(x_0)^{-n}\,\overline{z}\cosh(x_0)}{2\,k^2\sinh(x_0)^3} + \frac{z^2 W(x_0)^{-n}\,\overline{z}^2\cosh(x_0)\frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2\sinh(x_0)^3}$$
$$+ \frac{n z^2 W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} + \frac{n W(x_0)^{-n}\,\overline{z}^2}{2\,k^2\cosh(x_0)\sinh(x_0)} + \frac{n z^2 W(x_0)^{-n}\,\overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} - \frac{n z W(x_0)^{-n}\,\overline{z}\cosh(x_0)\frac{\partial}{\partial x_0} W(x_0)^2}{4\,k^2 W(x_0)^2\sinh(x_0)^3}$$
$$+ \frac{n z W(x_0)^{-n}\,\overline{z}\cosh(x_0)\frac{\partial^2}{(\partial x_0)^2} W(x_0)}{4\,k^2 W(x_0)\sinh(x_0)^3} + \frac{n^2 z W(x_0)^{-n}\,\overline{z}}{2\,k^2\cosh(x_0)\sinh(x_0)^3} + \frac{n^2 W(x_0)^{-n}\cosh(x_0)}{4\,k^2\sinh(x_0)^3} - \frac{z^2 W(x_0)^{-n}\frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)^2}$$
$$- \frac{W(x_0)^{-n}\,\overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)^2} - \frac{z^2 W(x_0)^{-n}\,\overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} + \frac{z W(x_0)^{-n}\,\overline{z}\cosh(x_0)\frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\sinh(x_0)^3} + \frac{z^2 W(x_0)^{-n}\frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)}$$
$$+ \frac{W(x_0)^{-n}\,\overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{z^2 W(x_0)^{-n}\frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n}\,\overline{z}^2 \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{n W(x_0)^{-n}\frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0)\sinh(x_0)^2}$$
$$+ \frac{n z W(x_0)^{-n}\,\overline{z}\frac{\partial}{\partial x_0} W(x_0)^2}{4\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} - \frac{n z W(x_0)^{-n}\,\overline{z}\frac{\partial^2}{(\partial x_0)^2} W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)^3} - \frac{3\,z^2 W(x_0)^{-n}\,\overline{z}^2}{16\,k^2\cosh(x_0)^3\sinh(x_0)^3} - \frac{n^2 W(x_0)^{-n}}{4\,k^2\cosh(x_0)\sinh(x_0)^3}$$
$$- \frac{z^2 W(x_0)^{-n}\frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} - \frac{W(x_0)^{-n}\,\overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} - \frac{z W(x_0)^{-n}\,\overline{z}\frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} + \frac{W(x_0)^{-n}\cosh(x_0)\frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2\sinh(x_0)^3}$$
$$+ \frac{n W(x_0)^{-n}\frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} - \frac{3\,z W(x_0)^{-n}\,\overline{z}}{8\,k^2\cosh(x_0)^3\sinh(x_0)^3} - \frac{W(x_0)^{-n}\frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} - \frac{3\,W(x_0)^{-n}}{16\,k^2\cosh(x_0)^3\sinh(x_0)^3} \tag{141}$$

$$
\begin{aligned}
\Phi'_{01} = {} & \frac{n^2 z^2 W(x_0)^{-n}\,\overline{z}\cosh(x_0)}{4\,k^2\sinh(x_0)^3} - \frac{n z^2 W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0}W(x_0)}{4\,k^2 W(x_0)\sinh(x_0)^2} - \frac{n^2 z^2 W(x_0)^{-n}\,\overline{z}}{4\,k^2\cosh(x_0)\sinh(x_0)^3} - \frac{n^2 z W(x_0)^{-n}\cosh(x_0)}{4\,k^2\sinh(x_0)^3} + \frac{z^2 W(x_0)^{-n}\,\overline{z}\cosh(x_0)\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\sinh(x_0)^3} \\
& + \frac{n W(x_0)^{-n}\,\overline{z}}{2\,k^2\cosh(x_0)\sinh(x_0)} + \frac{n z^2 W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} - \frac{n z W(x_0)^{-n}\cosh(x_0)\,\frac{\partial}{\partial x_0}W(x_0)^2}{8\,k^2 W(x_0)^2\sinh(x_0)^3} + \frac{n z W(x_0)^{-n}\cosh(x_0)\,\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,k^2 W(x_0)\sinh(x_0)^3} \\
& + \frac{n^2 z W(x_0)^{-n}}{4\,k^2\cosh(x_0)\sinh(x_0)^3} - \frac{W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0}W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)^2} - \frac{z^2 W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} + \frac{z W(x_0)^{-n}\cosh(x_0)\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\sinh(x_0)^3} \\
& + \frac{W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0}W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n}\,\overline{z}\,\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} + \frac{n z W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} - \frac{n z W(x_0)^{-n}\,\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)^3} \\
& - \frac{3\,z^2 W(x_0)^{-n}\,\overline{z}}{16\,k^2\cosh(x_0)^3\sinh(x_0)^3} - \frac{W(x_0)^{-n}\,\overline{z}\,\frac{\partial}{\partial x_0}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} - \frac{z W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} - \frac{3\,z W(x_0)^{-n}}{16\,k^2\cosh(x_0)^3\sinh(x_0)^3}
\end{aligned}
\tag{142}
$$

$$
\begin{aligned}
\Phi'_{02} = {} & \frac{n^2 z^2 W(x_0)^{-n}\cosh(x_0)}{4\,k^2\sinh(x_0)^3} - \frac{n z^2 W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)}{4\,k^2 W(x_0)\sinh(x_0)^2} - \frac{n^2 z^2 W(x_0)^{-n}}{4\,k^2\cosh(x_0)\sinh(x_0)^3} + \frac{z^2 W(x_0)^{-n}\cosh(x_0)\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\sinh(x_0)^3} \\
& + \frac{n W(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} + \frac{n z^2 W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} - \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)^2} - \frac{z^2 W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} \\
& + \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n}\,\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{3\,z^2 W(x_0)^{-n}}{16\,k^2\cosh(x_0)^3\sinh(x_0)^3} - \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2}
\end{aligned}
\tag{143}
$$

$$\Phi'_{10} = \frac{n^2 z W(x_0)^{-n} \overline{z}^2 \cosh(x_0)}{4\,k^2 \sinh(x_0)^3} - \frac{n z W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \sinh(x_0)^2} - \frac{n^2 z W(x_0)^{-n} \overline{z}^2}{4\,k^2 \cosh(x_0) \sinh(x_0)^3} - \frac{n^2 W(x_0)^{-n} \overline{z} \cosh(x_0)}{4\,k^2 \sinh(x_0)^3} + \frac{z W(x_0)^{-n} \overline{z}^2 \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3}$$
$$+ \frac{n z W(x_0)^{-n}}{2\,k^2 \cosh(x_0) \sinh(x_0)} + \frac{n z W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{n W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2 \sinh(x_0)^3} + \frac{n W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0) \sinh(x_0)^3}$$
$$+ \frac{n^2 W(x_0)^{-n} \overline{z}}{4\,k^2 \cosh(x_0) \sinh(x_0)^3} - \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \cosh(x_0)^2} - \frac{z W(x_0)^{-n} \overline{z}^2 \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3}$$
$$+ \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)} - \frac{z W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0) \sinh(x_0)} + \frac{n W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{8\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{n W(x_0)^{-n} \overline{z} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0) \sinh(x_0)^3}$$
$$- \frac{3\,z W(x_0)^{-n} \overline{z}^2}{16\,k^2 \cosh(x_0)^3 \sinh(x_0)^3} - \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{3\,W(x_0)^{-n} \overline{z}}{16\,k^2 \cosh(x_0)^3 \sinh(x_0)^3} \tag{144}$$

$$\Phi'_{11} = \frac{n^2 z W(x_0)^{-n} \overline{z} \cosh(x_0)}{4\,k^2 \sinh(x_0)^3} - \frac{n z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \sinh(x_0)^2} - \frac{n^2 z W(x_0)^{-n} \overline{z}}{4\,k^2 \cosh(x_0) \sinh(x_0)^3} - \frac{n^2 W(x_0)^{-n} \cosh(x_0)}{8\,k^2 \sinh(x_0)^3} + \frac{z W(x_0)^{-n} \overline{z} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3}$$
$$+ \frac{n z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{n W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3} + \frac{n W(x_0)^{-n} \cosh(x_0) \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{16\,k^2 W(x_0) \sinh(x_0)^3} \tag{145}$$
$$+ \frac{n^2 W(x_0)^{-n}}{8\,k^2 \cosh(x_0) \sinh(x_0)^3} - \frac{z W(x_0)^{-n} \overline{z} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} + \frac{W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{32\,k^2 W(x_0)^2 \sinh(x_0)^3} + \frac{n W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3}$$
$$- \frac{n W(x_0)^{-n} \frac{\partial^2}{(\partial x_0)^2} W(x_0)}{16\,k^2 W(x_0) \cosh(x_0) \sinh(x_0)^3} - \frac{3\,z W(x_0)^{-n} \overline{z}}{16\,k^2 \cosh(x_0)^3 \sinh(x_0)^3} - \frac{W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{32\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{3\,W(x_0)^{-n}}{32\,k^2 \cosh(x_0)^3 \sinh(x_0)^3}$$

$$\Phi'_{12} = \frac{n^2 z W(x_0)^{-n} \cosh(x_0)}{4\,k^2 \sinh(x_0)^3} - \frac{n z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{4\,k^2 W(x_0) \sinh(x_0)^2} - \frac{n^2 z W(x_0)^{-n}}{4\,k^2 \cosh(x_0) \sinh(x_0)^3} + \frac{z W(x_0)^{-n} \cosh(x_0) \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \sinh(x_0)^3} \tag{146}$$
$$+ \frac{n z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)}{8\,k^2 W(x_0) \cosh(x_0)^2 \sinh(x_0)^2} - \frac{z W(x_0)^{-n} \frac{\partial}{\partial x_0} W(x_0)^2}{16\,k^2 W(x_0)^2 \cosh(x_0) \sinh(x_0)^3} - \frac{3\,z W(x_0)^{-n}}{16\,k^2 \cosh(x_0)^3 \sinh(x_0)^3}$$

$$\Phi'_{20} = \frac{n^2 W(x_0)^{-n}\,\overline{z}^2\cosh(x_0)}{4\,k^2\sinh(x_0)^3} - \frac{nW(x_0)^{-n}\,\overline{z}^2\frac{\partial}{\partial x_0}W(x_0)}{4\,k^2 W(x_0)\sinh(x_0)^2} - \frac{n^2 W(x_0)^{-n}\,\overline{z}^2}{4\,k^2\cosh(x_0)\sinh(x_0)^3} + \frac{W(x_0)^{-n}\,\overline{z}^2\cosh(x_0)\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\sinh(x_0)^3}$$
$$+ \frac{nW(x_0)^{-n}}{2\,k^2\cosh(x_0)\sinh(x_0)} + \frac{nW(x_0)^{-n}\,\overline{z}^2\frac{\partial}{\partial x_0}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} - \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)}{4\,k^2 W(x_0)\cosh(x_0)^2} - \frac{W(x_0)^{-n}\,\overline{z}^2\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} \tag{147}$$
$$+ \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)^2}{8\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)} - \frac{W(x_0)^{-n}\,\frac{\partial^2}{(\partial x_0)^2}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)\sinh(x_0)} - \frac{3\,W(x_0)^{-n}\,\overline{z}^2}{16\,k^2\cosh(x_0)^3\sinh(x_0)^3} - \frac{W(x_0)^{-n}\,\frac{\partial}{\partial x_0}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2}$$

$$\Phi'_{21} = \frac{n^2 W(x_0)^{-n}\,\overline{z}\cosh(x_0)}{4\,k^2\sinh(x_0)^3} - \frac{nW(x_0)^{-n}\,\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{4\,k^2 W(x_0)\sinh(x_0)^2} - \frac{n^2 W(x_0)^{-n}\,\overline{z}}{4\,k^2\cosh(x_0)\sinh(x_0)^3} + \frac{W(x_0)^{-n}\,\overline{z}\cosh(x_0)\,\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\sinh(x_0)^3}$$
$$+ \frac{nW(x_0)^{-n}\,\overline{z}\frac{\partial}{\partial x_0}W(x_0)}{8\,k^2 W(x_0)\cosh(x_0)^2\sinh(x_0)^2} - \frac{W(x_0)^{-n}\,\overline{z}\frac{\partial}{\partial x_0}W(x_0)^2}{16\,k^2 W(x_0)^2\cosh(x_0)\sinh(x_0)^3} - \frac{3\,W(x_0)^{-n}\,\overline{z}}{16\,k^2\cosh(x_0)^3\sinh(x_0)^3} \tag{148}$$

$$\Phi'_{22} = \Phi_{22} \tag{149}$$

$$\Lambda' = \Lambda \tag{150}$$

References

[1] H. Stephani, D. Kramer, M. MacCallum, C. Hoenselaers, and E. Herlt, Exact Solutions of Einstein's Field Equations. Cambridge Monographs on Mathematical Physics, Cambridge University Press, 2009.

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